

SOCIOL 723: Social Statistics II

Week 9, Regression Discontinuity Designs

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Today

Sharp Regression Discontinuity

Fuzzy RD

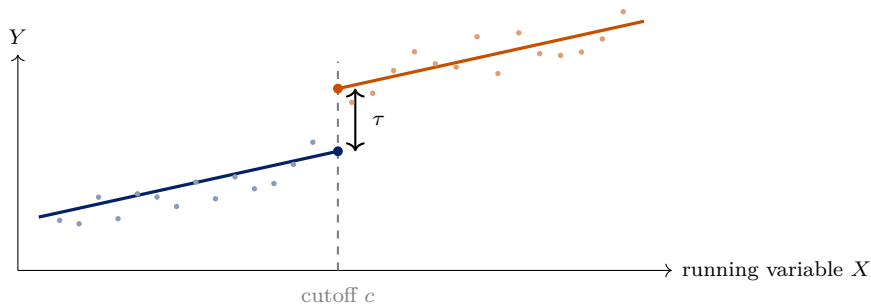
Validity

★ = essential, you must understand this

★ = for understanding, not examined

The Idea in One Picture ★

Many treatments are assigned by a rule: a test score threshold, an income cut-off, a vote share above 50%, an age of eligibility.



Just below and just above the cutoff, units are essentially identical, except that one group got the treatment. The jump in the outcome at c is the effect.

Sharp RD

Setup and Identification ★

A **running variable** (or forcing variable) X_i determines treatment deterministically:

$$D_i = \mathbf{1}\{X_i \geq c\}.$$

Continuity assumption (Hahn, Todd, and van der Klaauw 2001)

$E[Y_i(1) \mid X_i = x]$ and $E[Y_i(0) \mid X_i = x]$ are continuous in x at $x = c$.

Then

$$\tau_{RD} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x] = E[Y_i(1) - Y_i(0) \mid X_i = c].$$

- Note what is *not* assumed: nothing about the shape of $E[Y \mid X]$ away from the cutoff, and no ignorability anywhere else.

Setup and Identification ★ (cont.)

- **Positivity fails at every x , the cutoff included**: assignment is deterministic, so $\Pr(D = 1 \mid X = x)$ is 0 or 1 and never strictly between. Identification therefore does *not* come from overlap. It comes from **continuity** of $x \mapsto E[Y_i(d) \mid X_i = x]$, which lets the two one-sided limits stand in for the missing counterfactual at c . That is also why the estimand is local to c .

The Estimand Is Local, and That Matters ★

τ_{RD} is the average effect **for units at the cutoff**. Nothing more.

- A scholarship RD tells you about students at the eligibility margin, not about the strongest or weakest applicants.
- Extrapolating away from c requires a functional-form assumption of exactly the kind RD was designed to avoid.
- Compare with the LATE from Week 7: both are estimands you did not choose, determined by the design rather than by the question. State them plainly.
- In exchange, RD is often the most credible observational design there is. The identifying variation is visible in a scatterplot.

Lee (2008)'s reformulation is helpful: if agents cannot perfectly control X_i , then treatment near the cutoff is *as good as randomly assigned*, and the design is a local experiment.

Estimation: Local Linear Regression ★

Fit separate linear regressions on each side of the cutoff, using only observations within a bandwidth h , weighted by a kernel:

$$\hat{\tau} = \hat{\mu}_+ - \hat{\mu}_-, \quad (\hat{\mu}_+, \hat{\beta}_+) = \arg \min \sum_{i: c \leq X_i < c+h} K\left(\frac{X_i - c}{h}\right) (Y_i - \mu - \beta(X_i - c))^2,$$

and symmetrically below.

- **Local linear, not global polynomial.** High-order global polynomials give bizarre weights to distant observations and are now strongly discouraged (Gelman and Imbens 2019). Use degree 1 or 2.
- **Triangular kernel** (weights falling linearly to zero at h) is the standard choice and is boundary-optimal.
- Always let the slope differ on the two sides; forcing a common slope imposes a restriction you have no reason to believe.

The Estimator, Written Out

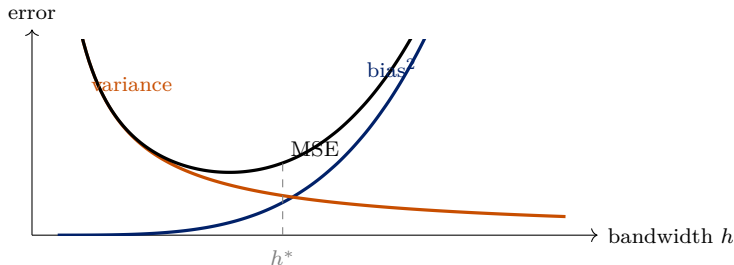
The RD estimate is the treatment coefficient in a **kernel-weighted, fully interacted local linear regression** on the observations within the bandwidth:

$$\min_{\mu, \tau, \beta_-, \beta_+} \sum_{i: |X_i - c| \leq h} K\left(\frac{X_i - c}{h}\right) \left(Y_i - \mu - \tau D_i - \beta_-(X_i - c)(1 - D_i) - \beta_+(X_i - c)D_i\right)^2,$$

with $D_i = \mathbf{1}\{X_i \geq c\}$ and $\hat{\tau}$ the estimate.

- **Centre the running variable at c** , so that the intercept shift τ is evaluated at the cutoff rather than at $X = 0$.
- **Interact the slope with D_i** . Forcing a common slope imposes a restriction you have no reason to believe, and it lets data on one side contaminate the estimate on the other.
- **Triangular kernel** $K(u) = (1 - |u|)_+$ weights observations linearly down to zero at the boundary and is minimax optimal at a boundary point. A uniform kernel is simpler and usually similar.

The Bandwidth Is the Whole Game ★



- **Wide h :** more data, lower variance, but the linear approximation is worse far from c , bias.
- **Narrow h :** less bias, but few observations, variance.
- Choose h by minimizing estimated MSE (Imbens–Kalyanaraman; the Calonico–Cattaneo–Titiunik version in `rdrobust`).
- The MSE-optimal bandwidth leaves a bias that invalidates the conventional interval. Use the CCT *robust bias-corrected* interval, which `rdrobust` reports by default.

Why the Optimal Bandwidth Breaks the Confidence Interval ★

The local linear estimator has an asymptotic expansion

$$\hat{\tau}_h - \tau \approx \underbrace{B h^2}_{\text{bias}} + \underbrace{O_p((nh)^{-1/2})}_{\text{noise}},$$

so squared bias grows like h^4 while variance falls like $1/(nh)$.

- Minimizing MSE gives $h^* \propto n^{-1/5}$, at which the bias and the standard error are **of the same order**.
- Consequently the conventional interval $\hat{\tau}_{h^*} \pm 1.96 \hat{se}$ is *not* centred on τ : it undercovers, sometimes badly.

Why the Optimal Bandwidth Breaks the Confidence Interval ★ (cont.)

- Calonico, Cattaneo, and Titiunik (2014): estimate the bias with a higher-order local polynomial, subtract it, and **widen the interval to account for having estimated the bias**. That is the “robust” row in `rdrobust` output.
- **Report the robust interval**. Reporting the conventional one at an MSE-optimal bandwidth is internally inconsistent.

Covariates, Power, and What to Report

- **Covariates are for precision only.** They must be *pre-determined*, they should enter additively (not interacted with D), and the estimate should barely move when you add them. If it moves a lot, the design is in trouble, not improved (Calonico et al. 2019).
- **Effective sample size:** report the number of observations *inside the bandwidth* on each side. An RD with $n = 100,000$ often has 400 observations doing the work, and the standard errors should be read accordingly.
- **Power:** RD is a local method and is inherently data-hungry. The required sample size is a substantial multiple of an experiment's for the same power, often quoted as around $3\times$, though the exact factor depends on the design, because we are extrapolating to a boundary from one side at a time.

Covariates, Power, and What to Report (cont.)

- **Discrete running variables and heaping:** rounded ages, coarse test-score grids, or income bunched at round numbers do not invalidate the design, but they do invalidate the continuous- X asymptotics behind bandwidth selection and bias correction. Use a discrete-score method and say so.

Fuzzy RD

When the Rule Is Not Followed ★

Often crossing the cutoff changes the *probability* of treatment rather than determining it: eligibility is not take-up.

$$\lim_{x \downarrow c} \Pr(D_i = 1 \mid X_i = x) - \lim_{x \uparrow c} \Pr(D_i = 1 \mid X_i = x) \in (0, 1).$$

The fuzzy RD estimator is a ratio of two sharp RD estimates:

$$\tau_{FRD} = \frac{\text{jump in } Y \text{ at } c}{\text{jump in } D \text{ at } c} = \frac{\lim_{x \downarrow c} E[Y \mid x] - \lim_{x \uparrow c} E[Y \mid x]}{\lim_{x \downarrow c} E[D \mid x] - \lim_{x \uparrow c} E[D \mid x]}.$$

- This is exactly the Wald estimator of Week 7, with $\mathbf{1}\{X \geq c\}$ as the instrument, estimated locally.
- The numerator alone is the **intention-to-treat** effect of crossing the cutoff, and it is worth reporting on its own, it needs no first-stage assumption.

Fuzzy RD Inherits Everything From IV ★

- **Exclusion:** crossing the cutoff must affect Y only through D . If the threshold also triggers something else, a different school, a different caseworker, a label, exclusion fails.
- **Monotonicity:** no one is treated *because* they fell below the cutoff.
- **The estimand is doubly local:** the effect for compliers at the cutoff. Not the ATE, not the ATT, and not the effect for everyone near the cutoff.
- **Weak first stage** is a real risk: if take-up jumps by only a few percentage points, you are dividing by a small noisy number and all the Week 7 pathologies return. Report the size of the jump in D .

Fuzzy RD Inherits Everything From IV ★ (cont.)

Related designs

Regression kink: identification from a change in *slope* rather than level.

Geographic RD: the running variable is distance to a boundary, and the “cutoff” is two-dimensional. **Multi-cutoff RD:** different units face different thresholds, which can be normalized and pooled.

Validity

The Threat: Manipulation ★

Continuity fails if units can *precisely* control which side of the cutoff they land on, retaking a test until you pass, reporting income just under a threshold, a firm holding hiring just below a regulatory headcount.

- **Imprecise control is fine.** Lee (2008)'s insight is that if agents cannot control X to the last decimal, assignment near the cutoff is as good as random even though agents influence X .
- **Density test** (McCrary 2008; Cattaneo, Jansson, and Ma 2020): is the density of X continuous at c ? A spike on the advantaged side is direct evidence of sorting.
- **Covariate smoothness:** run the RD with *pre-determined covariates* as outcomes. They should show no jump. This is the single most persuasive check, and it is the RD analogue of a balance table.
- **Donut hole:** drop observations immediately around c , where manipulation would concentrate, and see whether the estimate survives.

Other Checks and Common Errors

- **Placebo cutoffs:** estimate at fake thresholds away from c , using only data on one side so the real discontinuity is not picked up. Significant “effects” there mean your procedure finds jumps everywhere.
- **Bandwidth sensitivity:** plot $\hat{\tau}$ against h over a wide range with robust intervals. A result that exists only at one bandwidth is not a result.
- **Do not use a high-order global polynomial** (Gelman and Imbens 2019). The implied weights on distant observations are large, oscillating, and indefensible; estimates wander with the degree.
- **Do not control for post-treatment covariates**, and do not interact covariates with D unless you mean to estimate heterogeneity.
- **Do not report only the conventional interval** at an MSE-optimal bandwidth.

Reading and Reporting an RD ★

1. **The picture first.** Binned means of Y against X , with the fitted local polynomials and the cutoff. If the jump is not visible, be sceptical of the number.
2. The bandwidth, how it was chosen, and the effective n on each side.
3. Robust bias-corrected confidence intervals.
4. The density test and the covariate-smoothness checks.
5. A bandwidth-sensitivity plot and placebo cutoffs.
6. For fuzzy designs, the size of the jump in treatment take-up, and the intention-to-treat effect alongside.
7. An explicit statement that the estimand is local to the cutoff.

Looking Ahead

- RD buys extraordinary credibility at the cost of a very local estimand. The identifying variation is visible in a scatterplot, and the main threats (sorting, bandwidth dependence) are checkable.
- Note the family resemblance: sharp RD is a local experiment; fuzzy RD is local IV. The tools recombine rather than multiply.
- **Next week:** designs that use variation over *time* rather than across a threshold.
- Note again how the estimand was chosen by the design rather than by us. Saying so plainly is part of reporting an RD honestly.

Problem sets

Problem Set 3 is due Monday October 19. Problem Set 4 will be assigned next Tuesday (October 27) and is due Monday November 9; it covers Weeks 7 and 9.

References

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